

# Richard Cai

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## Education

**Cornell University, College of Engineering**

Ithaca, NY

B.S. Computer Science

2028

**Courses:** Algorithms, OS, Compilers, Systems/OO/Functional Programming, Machine Learning, Graphics, Digital Logic

## Technical Skills

- *Languages:* Python, Java, C, C++, JavaScript, Rust, OCaml, Assembly (x86/RISC-V)
- *Systems:* Linux, Git, Shell, Docker, CI/CD, Verilog (FPGAs)

## Experience

**University of Science and Technology of China | FuSeP 2026**

*Prof. Jiahui Hu, 2026*

- Developed AI quantization protection and circumvention measurement framework in adversarial image tasks across 15 attacks, 4 defenses, 6 models, and 9 quantizations. Presented poster to faculty and student audience.
- Implemented novel ablation/sweep fine-tuning, layerwise quantization and back-transfer attacks.

**Cornell | Research Assistant**

*Prof. Peter Liu, 2025-2026*

- Built a 25,000+ transaction commercial real estate database with custom scraping and deduplication pipeline, eliminating costly third-party data sources and enabling downstream ML modeling.
- Architecting end-to-end data pipeline from collection through AI analysis for a theater revenue prediction study.
- Developed high-precision large-scale newspaper filtering system in Linux combining temporal and textual signals.

**Cornell | CTECH YouTube Digital Twin**

*Dr. Cafer Avci, 2025*

- Built a YouTube Data API backend and multi-agent LLM framework to simulate creator behavior for digital twin research, enabling behavioral modeling at a scale infeasible with human subjects alone.

**Georgia Tech | Neurological Imaging Research**

*Prof. Jing Li, 2024*

- Researched pre-training and transfer learning techniques improving segmentation accuracy on brain MRI datasets.

**University of Washington | TEXTILE Program**

*Prof. Elizabeth Nance, 2023*

- Performed immunofluorescent cell imaging and computational analysis on microglia samples as part of an interdisciplinary bioengineering research program.

## Projects

**Eta-Haskell Java-based Compiler**

*Java, JFlex, CUP (yacc), x86 Assembly*

- Led a 4-person team as PM to deliver a 50,000+ LOC end-to-end compiler in 3 months, managing task breakdown, code review, and integration across all pipeline stages on deadline.
- End-to-end lexing, parsing, type checking, intermediate representation, assembly, and machine code generation.

**Facial Recognition App**

*JavaScript, React, JSON, Python, Face\_Recognition*

- Built a real-time facial recognition system with a React frontend and threaded Python REST backend.
- Implemented scalable face management, privacy controls, and a custom UI optimized for low-latency inference.

**Resource Export and HDI Analysis**

*Python, Scikit-learn*

- Led a team applying neural networks, random forests, decision trees, and K-means clustering to evaluate the "resource curse" hypothesis, finding strong groupings of countries with distinct export-to-HDI patterns.

**Report Parser Visualizer**

*Java, JavaScript, React, HTML, CSS*

- Developed Java backend to extract thousands of dynamic data points from daily Word logs using Google APIs.
- Built a cloud-deployed React frontend for interactive visualization of parsed report data.

**Safety-Net HDL Compiler**

*Rust*

- Implemented optimization passes and pattern transformations for a graduate research HDL compiler.